25900561 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY203 - INTRODUCTION TO OPTICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What is the main postulate of Planck's quantum theory? [K] / [CO1]
2. Which optical phenomena can be explained by Newton's corpuscular theory of light? [U] / [CO1]
3. Calculate the phase difference between two rays that differ by a path difference of λ/3 [U] / [CO2]
4. Define geometrical path length of light. [U] / [CO3]
5. Define a wavefront. Why is it called a phase front? [A] / [CO3]
6. What is the relation between the speed, frequency, and wavelength of light? [A] / [CO3]
7. What is a secondary wavefront? [U] / [CO4]
8. Write the lens-maker's formula for a lens of refractive index n placed in air. [U] / [CO4]
9. What is meant by a ray-transfer matrix? [U] / [CO4]

10. Distinguish between a thick lens and a thin lens. [U] / [CO4]

11. An optical system has a numerical aperture (NA) of 0.25 and uses light of wavelength λ=550 nm. Calculate the depth of focus. [A] / [CO6]

12. Write down the expression for f-number. [U] / [CO6]

13. Define the resolving power of a telescope. [U] / [CO6]

14. What is the purpose of an aperture in a camera? [U] / [CO6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Explain Maxwell's electromagnetic theory of light and state its key concepts. [U] / [CO1]

16. Light travels from glass (n=1.5) into water (n=1.33). If the wavelength of light in glass is 400 nm, determine: (a) the relative refractive index of water with respect to glass, (b) the wavelength of light in water. [A] / [CO3]

17. The critical angle for a medium is 60° when light passes from this medium into water (n = 1.33). Calculate the refractive index of the medium. What is the condition for total internal reflection in this case? [A] / [CO3]

18. Derive the expression for the translation matrix of a paraxial ray traveling through a homogeneous medium. Also, determine the value of the determinant of the translation matrix. [U] / [CO4]

19. An object is placed 15 cm in front of a thin lens and forms a real image 30 cm on the other side. What is the focal length of the lens? [A] / [CO4]

20. Derive the law of refraction (Snell's law) of a plane wave using Huygens' principle. [A] / [CO5]

21. Describe chromatic aberration in lenses and how one can reduce it? [U] / [CO6]

22. An astronomical telescope has an objective lens of focal length 10 m and an eyepiece of focal length 10 cm. Calculate the telescope's tube length and its magnification. [U] / [CO6]

[5x6 = 30]